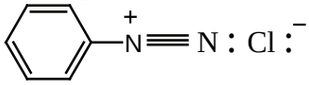
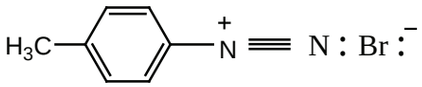
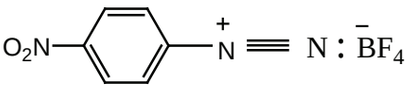
Example



Benzenediazonium Chloride



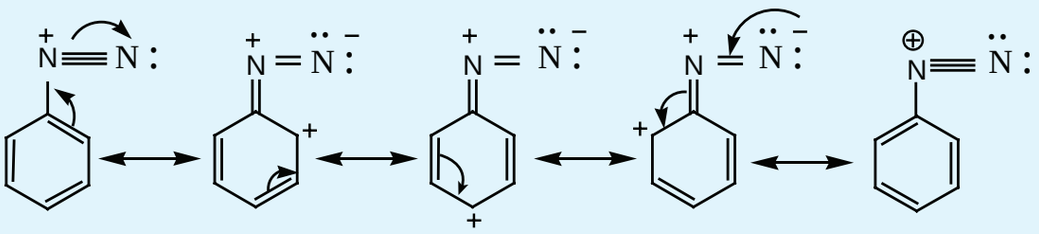
p - Toluenediazonium Bromide



p- Nitrobenzenediazonium tetra fluoroborate

13.3.2 Resonance structure

The stability of arene diazonium salt is due to the dispersal of the positive charge over the benzene ring.



Resonating structures of benzenediazonium salt

13.3.3 Method of preparation of Diazonium salts

We have already learnt that benzene diazonium chloride is prepared by the reaction of aniline with nitrous acid (Which is produced by the reaction of $NaNO_2$ and HCl) at 273 – 278K

13.3.4 Physical properties

- Benzene diazonium chloride is a colourless, crystalline solid.
- These are readily soluble in water and stable in cold water. However it reacts with warm water.
- Their aqueous solutions are neutral to litmus and conduct electricity due to the presence to ions.
- Benzenediazonium tetrafluoro borate is soluble in water and stable at room temperature.

13.3.5 Chemical reactions

Benzene diazoniumchloride gives two types of chemical reactions

A. Replacement reactions involving loss of nitrogen

In these reactions diazonium group is replaced by nucleophiles such as X^-,CN^-,H^-,OH^- etc.,

B. Reactions involving retention of diazogroup.

Coupling reaction.

XII U13-Organic Nitrogen Compounds.indd 219 11-01-2021 16:30:38